

Chapter 1

Snow basics

1.0.1 Direct and indirect methods for snow measurement

Knowledge of snow cover properties is essential for assessing solid water storage in the snowpack, a critical factor influencing agriculture and hydroelectric power generation, as well as for avalanche risk evaluation.

Historically, snow metrics were recorded manually, requiring an operator to conduct manual measurements on a daily basis whenever feasible. To ensure continuous data collection even in remote areas, automated weather stations (AWS) have been introduced, and are steadily replacing manual ones as well. An automated monitoring network provides a comprehensive overview for hazard assessment and for studying snow dynamics over decades. However, despite the significant number of AWS currently in operation, achieving high-resolution coverage remains challenging due to high costs and maintenance issues. Consequently, certain data can only be acquired via remote sensing, by means of Earth-observing satellites.

While remote sensing provides spatially distributed data across entire regions, manual and automated measures are taken at specific locations, which are chosen according to strict criteria. To ensure the reliability of these in situ observations, the monitoring site must match the features of the surrounding terrain. In alpine regions, monitoring sites must be clear of large boulders and protected from excessive sun exposure, which can cause premature snowmelt and yield unrepresentative data. It is also important to avoid locations prone to preferential snow accumulation driven by local wind patterns.

These guidelines represent only the baseline requirements for reliable snow measures. A more comprehensive analysis of the operational details is provided in the following sections.

Manual measures

Manual snow depth (HS) records are an effective tool for assessing long-term trends, as they can span over several decades. Their longevity is due to a straightforward measuring procedure, requiring only a stake or an extendible graduated rod.

The snow stake should be vertical, painted white to minimize melt and fixed to the ground before accumulation begins. It is required to make snow depth observations from a sufficient distance to avoid any deformation of the snowpack. Because the stake acts as an obstacle to wind and snow redistribution, localized mounds can form around it. Under such circumstances, operator must visually estimate the undisturbed snow surface to recover the actual HS value.

For a non-fixed observation, the snow depth value should be evaluated at the same location each time. The observer needs to ensure that the device, manually inserted in the snowpack, reaches the base surface without penetrating any soil thus overestimating the actual HS. The rod has to be vertical, and an estimate of the slope angle should be paired to every measure.

Closely linked to snow depth is snowfall depth (HN), which can only be obtained manually. To avoid wind-induced bias this measurement should be carried out in a sheltered location, so that snow accumulates undisturbed on an artificial surface for the prescribed interval (usually 24 hours). Just before clearing the surface, thereby starting the next observation period, a ruler is inserted vertically into the freshly formed layer to obtain a depth measurement.

Although HS and HN define the vertical extension of the snowpack, hydrological applications rely primarily on the snow water equivalent (SWE). As shown in (cite equation), SWE is directly linked to snow depth through the bulk density of the snowpack. Although various manual measurement techniques exist, this section focuses on the two primary methods, snow pits and snow tubes.

A snow pit consists of a manually-dug hole up to a reference surface, usually the ground. Initially, the observer performs a stratigraphic analysis to deepen his knowledge of the internal structure of the snowpack. The actual SWE measurement is conducted by inserting a sampler of known volume perpendicularly into the exposed snow face; the sample is then weighted to assess snow density. This procedure is usually repeated at multiple depths, either for each identified layer or at fixed intervals, such as every 20 centimetres. This method enables the observer to reconstruct the snow density profile which, when combined with depth information, yields both the layer-by-layer and total snow water equivalent at that specific location. Clearly, the

snow pit technique is disruptive and needs to be performed at a different site each time.

If the measurement campaign is focused on a spatial survey rather than a detailed, locally defined inspection, the logical choice is utilizing a snow tube, because it allows for rapid sampling of the entire snow column to directly retrieve the total SWE, sacrificing layer-specific information. The snow tube is inserted vertically in the snowpack until it reaches the ground. An estimate of the extracted core volume is made, and the average density is then computed by weighting the snow column. The conversion to SWE is then trivial. This device should be used where the snow water equivalent is less spatially homogeneous, such as locations with heterogeneous exposure to both wind patterns and solar radiation.

Automatic measures

Automatic weather station (AWS) networks do not require manual intervention, except for routine maintenance and instrument calibration. An AWS provides continuous observations even in remote locations where human operation would be unfeasible. Typically, an alpine automatic weather station is equipped with a variety of sensors to provide a highly comprehensive overview of environmental conditions.

Automated HS measurements can be conducted using instruments based on either sonic or optical (laser) technology. Sonic sensors transmit an ultrasonic pulse towards the snow surface and detect the returning echo. Since the speed of sound is temperature-dependent, a temperature reading is also required to ensure accuracy. On the other hand, laser instruments determine the target surface distance by analyzing signal interference. These measurements are actually much more precise, but they come at a higher energy cost. In both scenarios, the sensor measures the distance to the upper layer of the snowpack, from which the snow depth is then retrieved

$$\text{HS} = \text{Zero depth distance} - \text{Distance to target} \quad (1.1)$$

The zero depth distance must be determined before snow accumulation begins, and is a key part of instrument calibration. To obtain reliable measurements, the HS sensor should be installed at a sufficient height above the maximum anticipated snow depth to prevent burial during winter, which would result in data loss. In addition, minimizing interference with the support structure is essential, as the mast acts as a wind obstacle and can induce measurement artifacts. Consequently, HS sensors are typically installed on a horizontal arm that stretches outward, ensuring that the target area remains at a sufficient distance from the mounting pole.

SWE monitoring can also be automated; to this end, several techniques have been developed, the most common of which relies on snow pillows. This instrument consists of a rubber bladder filled with an antifreeze liquid mixture and coupled with a pressure transducer, which provides the SWE estimate. The snow pillow is installed flush with the ground surface, thereby supporting the weight of the entire snowpack. As snow accumulates on top of the pillow, the hydrostatic pressure increases and this variation is transduced into snow water equivalent values. Snow pillows were first introduced in the late 1960s, and have proved to be a useful tool for SWE measurements (Beaumont 1965). However, they are currently being replaced by snow scales due to the toxicity of the antifreeze fluid, which poses an environmental hazard to the installation site.

Remote sensing

Satellite remote sensing (SRS) serves as a useful tool to fill critical gaps in current knowledge regarding climatic variables, enabling researchers to monitor, usually on a daily basis, even remote locations which could not be accessible by a human operator. Furthermore, SRS data substantially improve meteorological reanalysis products, which constitute a cornerstone of contemporary climate change research. Earth observing satellites can be classified into two distinct categories based on their orbital characteristics:

- ▶ Polar orbit satellites, which travel roughly from north to south over the poles, targeting a specific longitudinal strip per pass. The entire globe is thus mapped over consecutive orbits. Usually, these satellites operate at an altitude of approximately 700 km, completing a full revolution in under two hours.
- ▶ Geostationary satellites, which occupy a geosynchronous orbit, thereby allowing for real-time meteorological monitoring on a specific target area. Their altitude of approximately 36000 km above Earth's equator results in a significantly lower spatial resolution compared to that of polar orbit satellites.

The choice of orbit is strictly driven by observational priorities. Consequently, satellites dedicated to cryospheric research favor polar orbit, as maximizing spatial resolution is essential to capture the highly variable distribution of snow coverage across complex terrains.

Satellite-based observations of Snow Cover Extent (SCE) began in 1967, through the efforts of National Oceanic and Atmospheric Administration (NOAA). This monitoring program focused on the Northern Hemisphere, where most of the world's

terrestrial snow cover is found, with observations conducted on a weekly basis. Despite the preliminary nature of this work, it highlighted the strong latitudinal dependence of snow cover duration and paved the way for more comprehensive analyses. One of the main limitations of these early observations was their coarse spatial resolution of ~ 200 km, which was insufficient to adequately characterize snow cover in many mountainous regions (Bormann et al. 2018).

The next big leap in sensor technology came with the introduction of the Advanced Very High Resolution Radiometer (AVHRR), which measures Earth radiance in five spectral bands to obtain information on temperature, albedo and cloud coverage distribution. The first tests on a spaceborne sensor took place in 1978, but reliable snow cover data are available only from 1981 onward. This snow product has been used to estimate snow cover duration and trends since the middle eighties in different mountainous regions such as the Alps, where SRS data highlight significantly negative SC trends throughout the 1990s, followed by a partial recovery at the beginning of the new millennium (Hüsler et al. 2014). Although AVHRR offers a significant improvement in spatial resolution with its pixels of ~ 1 km, data reliability is limited by the presence of only one thermal channel for snow-cloud discrimination (Fugazza et al. 2021).

Chapter 2

Snow products

2.1 IT-Snow

IT-SNOW is a high resolution, multi-year snow reanalysis dataset covering the whole Italian territory. It provides daily maps of snow water equivalent (SWE), snow depth, bulk snow density and liquid water content of the snowpack for a period spanning from September 1st, 2010 to August 31st, 2025. The spatial resolution is approximately 500 meters. IT-SNOW maps are the output of an operational chain developed by CIMA Research Foundation called S3M Italy.

2.1.1 S3M

S3M, also known as *Snow Multidata Mapping and Modelling*, is a hydrology-oriented cryospheric model that simulates seasonal snow evolution through time. S3M is a raster-based model treating pixels as independent units, as it neglects mass and energy transport processes such as wind redistribution. The same set of ordinary differential equations (ODEs) is solved for each grid point by means of the Euler method: the hourly time-step can be adjusted by the user to meet specific application needs. All input and output variables are distributed.

S3M is specifically designed for large-scale, real-time application in order to be an useful tool in flood forecasting frameworks. This model achieves a high level of performance by focusing on key snow evolution drivers via precipitation-phase partitioning and a hybrid temperature and radiation-driven melt parametrization.

2.2 MODIS

The role of Satellite Remote Sensing (SRS) in global climate monitoring has become increasingly critical, providing spatio-temporal observations across remote regions where conventional ground-based methods are lacking. In particular, those kinds of measurements are highly effective for mapping snow cover extent and duration. NASA’s Moderate Resolution Imaging Spectroradiometer (MODIS) is a major tool in this field: this 36-band sensor operates aboard two Earth Observing System (EOS) satellites, namely Terra (launched in December 1999) and Aqua (launched in May 2002). This tool has been used in order to assess snow cover duration (SCD) trends across the Greater Alpine Region (Fugazza et al. 2021) and to establish SCD climatologies across Europe. One of the latest studies on snow cover metric characteristics and trends focuses on the Italian territory (Manara et al. 2025), and therefore on the Italian Alps, which are the primary focus of this thesis.

2.2.1 Dataset Generation and Gap-Filling Strategy

In its latest version, this Italian snow cover dataset spans from 2000 to 2025 and consists of binary maps, where a value of zero represents to bare ground, whereas one is associated only to snow covered pixels. Available on a daily basis, these maps feature a spatial resolution of roughly 500 meters.

The starting point of the workline that led the authors to those maps are the raw modis datas. To correctly identify snow covered pixels advantage is taken on a property of snow, which is that it has very high reflectance in the visible part of the electromagnetic spectrum, whereas it absorbs most of the shortwave radiation. This characteristics is highlighted by the normalized difference snow index (NDSI), which is defined as

$$\text{NDSI} = \frac{\text{band4} - \text{band6(7)}}{\text{band4} + \text{band6(7)}} \quad (2.1)$$

where band 4 is in the visible range (in particular $0.545 - 0.565 \mu\text{m}$) and band 6 (or 7) is in the shortwave infrared (namely $1.628 - 1.672 \mu\text{m}$ for Terra and $2.105 - 2.155 \mu\text{m}$ for Aqua).

2.3 Po Basin

This snow dataset focuses on the mountainous part of the Po River District, which is an over-regional entity resulting from the union of the Po River Basin with some

smaller ones such as Conca and Reno basins. This territory, spanning eight regions of Northern Italy and parts of Switzerland and France, covers almost 87000 km^2 . Assessing the amount of water stored as snow in the highlands of this region is particularly important, because the Po River Basin is home to roughly one-third of Italy's total population and produces almost 55% of Italy's hydropower. This snow product is a thirty years long SWE reconstruction, covering a period spanning from 1992 to 2021. For every hydrological year, SWE maps for the mountainous part of the Po River District are given on a daily basis during the period spanning from October 3rd to July 1st, which roughly corresponds to the period where snow coverage extends beside glacierized area. These SWE maps have a spatial resolution of $500 \times 500\text{ m}^2$ and are created using a hybrid modeling approach "J-snow".

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